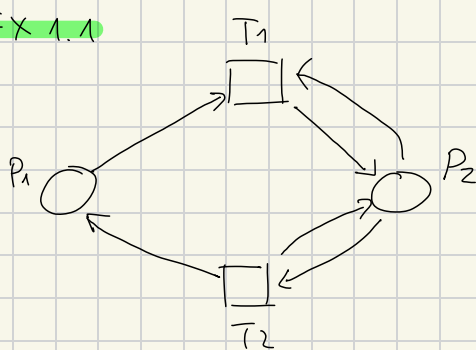
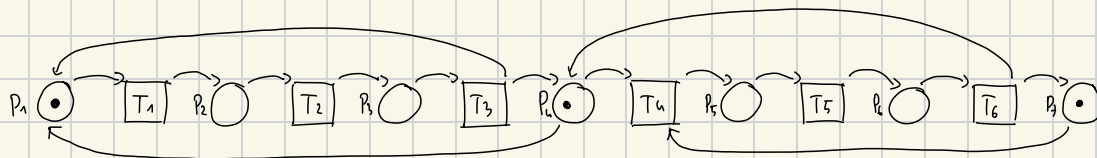


EX 1.1



EX 1.2



EX 1.3

$$P = \{P_1, P_2\}$$

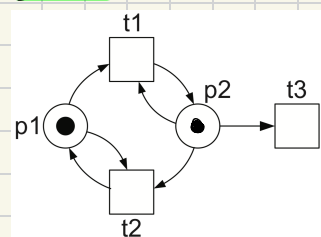
$$T = \{T_1, T_2, T_3\}$$

$$F = \{(P_1, T_1) (P_1, T_2) (T_1, P_2) (P_2, T_1) (P_2, T_3) (P_2, T_2) (T_2, P_1)\}$$

$$\begin{aligned} \bullet T_1 &= \{P_1, P_2\} & \bullet T_2 &= \{P_1, P_2\} & \bullet T_3 &= \{P_2\} \\ T_1^* &= \{P_2\} & T_2^* &= \{P_1\} & T_3^* &= \emptyset \end{aligned}$$

We don't have enable transitions.

EX 1.4



$$\bullet T_1 = \{P_1, P_2\}$$

$$T_1^* = \{P_2\}$$

$$\bullet T_2 = \{P_1, P_2\}$$

$$T_2^* = \{P_1\}$$

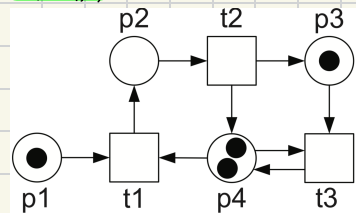
$$\bullet T_3 = \{P_1\}$$

$$T_3^* = \{P_2\}$$

$$\begin{aligned} 1) & \text{ All } T_1, T_2, T_3 \\ 2, 3, 4) & [P_1, P_2] \\ & \quad \swarrow \quad \searrow \\ & [P_2] \quad [P_1] \\ & \quad \swarrow \quad \searrow \\ & [P_1] \quad [P_2] \end{aligned}$$

5) $[P_1, P_2], [P_2], [P_1], []$ are reachable markings
 $[P_1], []$ are terminal markings

EX 1.5



$$1) P = \{P_1, \dots, P_4\}$$

$$T = \{T_1, T_2, T_3\}$$

$$F = \{(P_1, T_1) (T_1, P_2) (P_2, T_2) (T_2, P_3) (T_2, P_4) (P_3, T_3) (T_3, P_4) (P_4, T_1) (P_4, T_2)\}$$

$$m_0 = [P_1, P_3, P_4^2]$$

$$2) \bullet T_1 = \{P_1, P_4\} \quad \bullet T_2 = \{P_1\} \quad \bullet T_3 = \{P_3, P_4\}$$

$$T_1^* = \{P_2\}$$

$$T_2^* = \{P_3, P_4\}$$

$$T_3^* = \{P_4\}$$

$$3) T_1, T_3$$

$$[P_1, P_3, P_4^2]$$

$$\swarrow T_1 \quad \searrow T_3$$

$$[P_2, P_3, P_4]$$

$$[P_1, P_4^2]$$

$$\swarrow T_2 \quad \searrow T_3$$

$$[P_2^2, P_4^2]$$

$$[P_1, P_4]$$

$$\swarrow T_2 \quad \searrow T_3$$

$$[P_4^2]$$

$$[P_3, P_4^2]$$

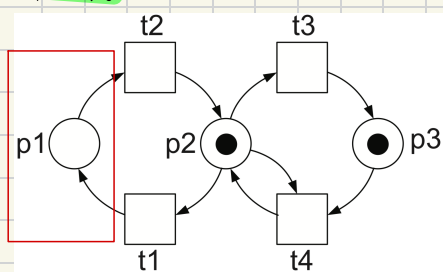
$$4) \text{ REACHABLE: } [P_1, P_3, P_4^2] [P_2, P_3, P_4] [P_1, P_4^2] [P_2^2, P_4^2] [P_2, P_4] [P_4^2] [P_3, P_4^2]$$

$$5) \text{ TERMINAL: } [P_4^2]$$

$$6) \text{ Yes } [P_1, P_3, P_4^2] [P_2, P_3, P_4]$$

$$7) \text{ No}$$

Ex 1.6



$$1) P = \{p_1, p_2, p_3\}$$

$$T = \{T_1, T_2, T_3, T_4\}$$

$$F = [(p_1, T_2)(T_1, p_2)(T_2, p_3)(p_3, T_4)(T_3, p_1)(p_2, T_3)(T_4, p_2)]$$

$$m_0 = [p_2, p_3]$$

$$2) \cdot T_1 = \{p_2\} \quad \cdot T_2 = \{p_1\} \quad \cdot T_3 = \{p_2\} \quad \cdot T_4 = \{p_2, p_3\}$$

$$T_1^* = \{p_1\} \quad T_2^* = \{p_2\} \quad T_3^* = \{p_3\} \quad T_4^* = \{p_2\}$$

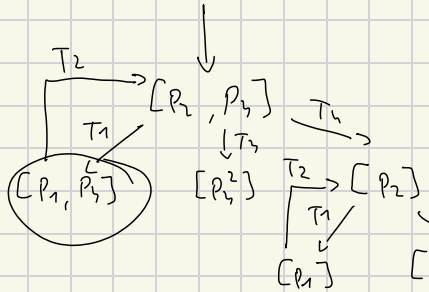
$$3) T_1, T_3, T_4$$

$$4) [p_2, p_3][p_1, p_3][p_3^1][p_2, p_1][p_3]$$

$$5) [p_3][p_3^1]$$

$$6) [p_2, p_3][p_2]$$

7) Increases infinitely



Ex 1.7

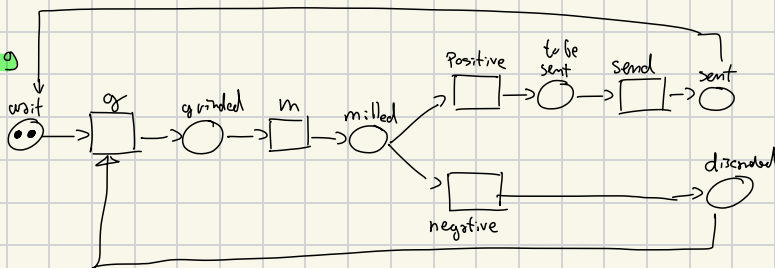
$$m'(p) = \begin{cases} m(p) - 1, & \text{if} \\ m(p) + 1, & \text{if} \\ m(p), & \text{if} \end{cases}$$

a transition T has P in his preset and not in the postset

a transition T has P in his postset and not in the preset

a transition T has P in his postset and in his preset

Ex 1.9



Ex 1.10

